Wyndor Glass – Graphical Method

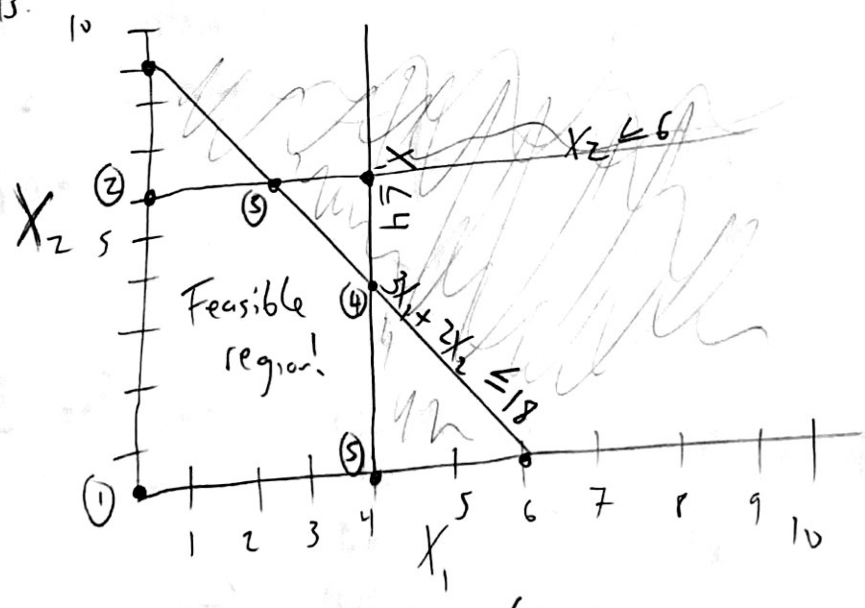
$$\text{Max}(z) = 3X_1 + 5X_2$$

subject to:

$$X_1 \le 4 \quad (1)$$
$$2X_2 \le 12 \quad (2)$$
$$3X_1 + 2X_2 \le 18 \quad (3)$$

Solve for intersection of constraints with axes, draw the graph, identify feasible region, solve for corner point coordinates, plug and chug in objective function!

The first two constraints are easy to draw because they are constants.



- $X_1 \le 4$ is a vertical line at $(4, 0)$
- $2X_2 \le 12$ is a horizontal line, but you need to simplify.

  $X_2 \le \frac{12}{2}$ becomes $X_2 \le 6$ which is easily plotted.
- now solve for third constraint, when $X_1 = 0$, $X_2 = 9$

  when $X_2 = 0$, $X_1 = 6$

not sure where to shade? try a set of coords!

We can see five potential corner points. Some can be read easily from the graph

C Point 1 = $(0, 0)$ C Point 5 = $(4, 0)$

C Point 2 = $(0, 6)$ C Point 3 = $(2, 6)$ and CP is $(4, 3)$

but CP 3 and 4 require some math.

- CP 3 is the intersection of Eqn 2 and Eqn 3.

$$\begin{array}{rr} 3X_1 + 2X_2 = 18 \\ - \quad 2X_2 = 12 \\ \hline 3X_1 = 6 \end{array}$$

just solve for X_2
no need for multiplication

$$\boxed{X_1 = 2}$$

now plug into one of the original constraints (equations) and solve for X_2.

$$3(2) + 2(X_2) = 18$$

$$\boxed{X_2 = \frac{(18-6)}{2} = 6}$$

- CP 3 is $(2, 6)$

- CP4 is the intersection of Eqn 1 and Eqn 3

$$\begin{array}{rr} 3X_1 + 2X_2 = 18 \\ 3X_1 \quad = 3(4) \\ \hline 2X_2 = 6 \end{array}$$

← times 3!

$$\boxed{X_2 = 3}$$

now plug and chug

$$3X_1 + (2)(3) = 18$$

$$\boxed{X_1 = \left(\frac{18-6}{3}\right) = 4}$$

- CP 4 is $(4, 3)$

(now you can label the plot)

Now evaluate the objective function $3X_1 + 5X_2$

$CP1 = (0,0) = 3(0) + 5(0) = 0$

$CP2 = (0,6) = 3(0) + 5(6) = 30$

$CP3 = (2,6) = 3(2) + 5(6) = 36$

$CP4 = (4,3) = 3(4) + 5(3) = 27$

$CP5 = (4,0) = 3(4) + 5(0) = 12$

$\therefore$ CP3 maximizes the objective function when $X_1 = 2$ and $X_2 = 6$ and results in $Z = 36$,

Done!

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Wyndor Glass: Simplex Geometric Interpretation

Here's the original problem statement:

$$\text{Max}(Z) = 3X_1 + 5X_2$$

subject to:

$$\begin{aligned} X_1 &\le 4 \\ 2X_2 &\le 12 \\ 3X_1 + 2X_2 &\le 18 \end{aligned}$$

First things first, write the problem in "augmented form"

$$\text{Max}(Z) = 3X_1 + 5X_2$$

subject to:

$$\begin{aligned} X_1 + X_3 &= 4 \\ 2X_2 + X_4 &= 12 \\ 3X_1 + 2X_2 + X_5 &= 18 \end{aligned}$$

what is the initial basic solution? $(0, 0, 4, 12, 18)$

- $(0,0)$ is always a nice initial BF (basic feasible) solution
- X_1 and X_2 are nonbasic (zero) while X_3, X_4, X_5 are basic (zero)

- We know that we will move to an adjacent feasible solution along the edge of the feasible region – but which one?

- look at the objective function! The slopes (rates of improvement) are given.

  $X_1 = 3$ vs. $X_2 = 5$... let's go up X_2! but how far?

To answer the question of how far to go, we need to conduct the minimum ratio test. (2

Let's write our functional constraints again — basic variables are underlined

(1) $X_1 \qquad + \underline{X_3} \qquad = 4$

(2) $\qquad 2X_2 \qquad + \underline{X_4} = 12$

(3) $3X_1 + 2X_2 \qquad + \underline{X_5} = 18$

(the basic variables are nonzero and have a coefficient of 1, and only appear once)

how far up X_2 can we go? we know $X_1 = 0$....
we rewrite in terms of basic variables

(1) $\underline{X_3} = 4$ means X_2 is unbounded

(2) $2X_2 + X_4 = 12$ becomes $\underline{X_4} = 12 - 2X_2 \geq 0$ so X_2
so $-2X_2 \geq -12$ which makes $X_2 \leq 6$

(3) $3X_1 + 2X_2 + X_5 = 18$ becomes $\underline{X_5} = 18 - 3X_1 - 2X_2 \geq 0$
so $-2X_2 \geq -18 + 3(0)$ which makes

$$X_2 \leq 9$$

(remember, when you divide by a negative number, switch the inequality)

This means we can up to $X_2 = 6$ and then stop.

X_2 was entering basic variable and
X_4 was leaving basic variable.

So, what do we know?

initial BF solution

nonbasic $X_1 = 0 \quad X_2 = 0$

basic $X_3 = 4, X_4 = 12, X_5 = 18$

See how nonbasic (zero) variables changed?

Iteration 1 BF sol'n

$X_1 = 0 \quad X_4 = 0$

$X_2 = 6$ (just solved)

$X_3 = ?$ and $X_5 = ?$

This is a great time Gaussian - Jordan elimination which makes it easy to read off the basic variables (appear once in each constraint w/ coefficient of 1)

let's write with the objective function (so we can rewrite as a function of new nonbasic variables X_1 and X_4

$$
\begin{array}{llllllll}
R(0) & Z & -3X_1 & -5\underline{X_2} & & & & = 0 \\
R(1) & & X_1 & & +\underline{X_3} & & & = 4 \\
R(2) & & & 2\underline{X_2} & & +X_4 & & = 12 \\
R(3) & & 3X_1 & +2\underline{X_2} & & & +\underline{X_5} & = 18
\end{array}
$$

our basic variables are underlined, X_2, X_3 and X_5,

let's clean up those coefficients by: (we are only cleaning up for readability)

(1) multiplying (dividing) by a constant

(2) add (subtract) a multiple of one row to another

$R_2^* = R_2/2$ gives $X_2 + \frac{1}{2}X_4 = 6$ (which makes sense!)

then, $R_0^* = R_0 + 5R_2^*$ $\quad R_3^* = R_3 - 2R_2^*$ (this is the limit we just calculated!!!)

Applying these operations gives

$R_0^* = R_0 + 5R_2$

(note that by doing this you've expressed the obj. fn in terms of nonbasic (zero) variables!)

$$
\begin{array}{rl}
 & Z - 3X_1 - 5X_2 = 0 \\
+ & 5\left(X_2 + \tfrac{1}{2}X_4 = 6\right) \\
\hline
 & Z - 3X_1 + \tfrac{5}{2}X_4 = 30
\end{array}
$$

hence, when $X_1 = 0$ and $X_2 = 6$, $Z = 30$. more on that in a minute.

$R_3^* = R_3 - 2R_2^*$

$$
\begin{array}{rl}
 & 3X_1 + 2X_2 + X_5 = 18 \\
- & 2\left(X_2 + \tfrac{1}{2}X_4 = 6\right) \\
\hline
\end{array}
$$

$R_3^* = 3X_1 - X_4 + X_5 = 6$

now let's rewrite our rows one more time for easy viewing

$$
\begin{array}{llcl}
R(0) & Z - 3X_1 + \tfrac{5}{2}X_4 & = & 30 \\
R(1) & X_1 + \underline{X_3} & = & 4 \\
R(2) & \underline{X_2} + \tfrac{1}{2}X_4 & = & 6 \\
R(3) & 3X_1 - X_4 + \underline{X_5} & = & 6
\end{array}
$$

underline the basic variables. $X_2 = 6$, $X_3 = 4$ and $X_5 = 6$

this BF of $Z = 30$ is given by $(0, 6, 4, 0, 6)$ ... but is it optimal?

... is $(x_1, x_2, x_3, x_4, x_5) = (0, 6, 4, 0, 6)$ optimal?

look at the objective function (new)

$$Z - 3x_1 + \frac{5}{2}x_4 = 30$$

$$Z = 30 + 3x_1 - \frac{5}{2}x_4$$

we can further improve the obj function!

Iteration 2

Now look at the slopes! x_1 has a positive 3... which means x_1 is the entering basic variable.

how much can we increase x_1? minimum ratio test

(1) $x_1 + \underline{x_3} = 4$

(2) $\underline{x_2} + \frac{1}{2}x_4 = 6$

(3) $3x_1 - x_4 + \underline{x_5} = 6$

let's rewrite in terms of basic variables. (x_2, x_3, x_5)

(1) $x_3 = 4 - x_1 \geq 0$ becomes $-x_1 \geq -4$

divide by -1 gives $x_1 \leq 4$

(2) $x_2 = 6 - \frac{1}{2}x_4 \geq 0 \Rightarrow$ unbounded for x_1

(3) $x_5 = 6 - 3x_1 + x_4 \geq 0$ which becomes $-3x_1 \geq -6$

divide by -3 gives $x_1 \leq 2$

x_5 is the leaving basic variable (will become nonbasic/zero)

$x_1 = 2$ is the max we can increase x_1.

now it's clean up time - GJ elimination!

$X_1 = 2$ is the entering basic variable.

$X_2 = 6 \quad X_4 = 0 \quad X_3 = ? \quad X_5 = 0$

$$
\begin{array}{llllll}
(0) & Z - \underline{3X_1} & & + \frac{5}{2}X_4 & & = 30 \\
(1) & \underline{X_1} & + \underline{X_3} & & & = 4 \\
(2) & \underline{X_2} & & + \frac{1}{2}X_4 & & = 6 \\
(3) & \underline{3X_1} & & - X_4 & + X_5 & = 6
\end{array}
$$

we need to get rid of the extra X_1 for easy readability.

- one basic variable per constraint w/ a coeff = 1

$$R_3^* = R_3/3 = X_1 - \frac{1}{3}X_4 + \frac{1}{3}X_5 = 2$$

$$R_1^* = R_1 - R_3^* = X_1 + X_3 - (X_1 - \frac{1}{3}X_4 + \frac{1}{3}X_5) = 4 - 2$$

$$R_1^* = X_3 + \frac{1}{3}X_4 - \frac{1}{3}X_5 = 2$$

$$R_0^* = R_0 + 3R_3^*$$

$$Z - 3X_1 + \frac{5}{2}X_4 + 3(X_1 - \frac{1}{3}X_4 + \frac{1}{3}X_5) = 30 + 3(2)$$

$$Z + \frac{5}{2}X_4 - \frac{3}{3}X_4 + \frac{3}{3}X_5 = 36$$

$$Z + \frac{15}{6}X_4 - \frac{6}{6}X_4 + X_5 = 36$$

$$Z + \frac{9}{6}X_4 + X_5 = 36$$

$$Z + \frac{3}{2}X_4 + X_5 = 36$$

of course, let's rewrite everything for clarity.

$$
\begin{array}{llllllll}
(0)^* & Z - & & & + \frac{3}{2}x_4 & + x_5 & = 36 \\
(1)^* & & & \underline{x_3} & + \frac{1}{3}x_4 & - \frac{1}{3}x_5 & = 2 \\
(2)^* & & \underline{x_2} & & + \frac{1}{2}x_4 & & = 6 \\
(3)^* & \underline{x_1} & & & - \frac{1}{3}x_4 & + \frac{1}{3}x_5 & = 2
\end{array}
$$

underline basic variables. (for clarity)

the solution $(x_1, x_2, x_3, x_4, x_5) = (2, 6, 2, 0, 0)$

and $Z = 36$.

now, optimality test — rewrite obj fn.

$$Z - \frac{3}{2}x_4 + x_5 = 36$$

$$Z = 36 - \frac{3}{2}x_4 - x_5$$

∴ do you see a positive slope? No!

You are optimal! write the solution.

$(2, 6, 2, 0, 0)$ //

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Wyndor - Simplex

(Given all LHS $\leq$ RHS, this is a standard simplex maximization problem.)

Original Form of the Model:

$$Max(Z) = 3X_1 + 5X_2$$

subject to:

$$\begin{aligned} X_1 \quad &\leq 4 \\ 2X_2 &\leq 12 \\ 3X_1 + 2X_2 &\leq 18 \end{aligned}$$

and $X_1, X_2 \geq 0$ (non-negativity)

Augmented form of the model:

(slack variables omitted from obj fn because coefficients are zero!)

$$Max(Z) = 3X_1 + 5X_2$$

subject to:

$$\begin{aligned} X_1 + X_3 &= 4 \\ 2X_2 + X_4 &= 12 \\ 3X_1 + 2X_2 + X_5 &= 18 \end{aligned}$$

and $X_1, X_2, X_3, X_4, X_5 \geq 0$ (non-negativity)

Now construct the initial simplex tableau

X_1	X_2	X_3	X_4	X_5	RHS
1	0	1	0	0	4
0	2	0	1	0	12
3	2	0	0	1	18
-3	-5	0	0	0	0 ← Z

(note: obj fn is now in bottom row of tableau!)

rewrite obj fn so Z is on the RHS.

$-3X_1 - 5X_2 = Z$ and initial BFS is $(0, 0, 4, 12, 18)$

which gives $Z = 0$

Now it's time to start an iteration - we rewrite the initial simplex tableau. (R_3 is the obj fn) (2)

$\square$ = pivot element (Make that a 1!)

<u>Iteration 1</u>

	X_1	X_2	X_3	X_4	X_5	RHS
R_0	1	0	1	0	0	4
R_1	0	$\boxed{2}$	0	1	0	12
R_2	3	2	0	0	1	18
R_3	-3	-5	0	0	0	0

X_2 is entering basic variable

X_4 is leaving basic variable

where do we start? choose most negative value in R_3 (bottom row, obj fn) then a minimum ratio test

- $R_0 = 4/0 = \text{Inf} \rightarrow$ unbounded
- $R_1 = 12/2 = 6 \rightarrow$ Minimum!
- $R_2 = 18/2 = 9$

- R_1 has the minimum b/a so X_2 can be increased to 6.
- $\therefore$ That means the 2 in the pivot element needs to become a 1, and all values above and below must become 0 (GJ elim).
- Here are the row operations:

$$R_1^* = \frac{1}{2}R_1 \quad \text{then} \quad R_2^* = R_2 - 2R_1^*$$

$$\text{and} \quad R_3^* = R_3 + 5R_1^*$$

Math on the next page.

$R_1^* = \frac{1}{2} R_1$ so $\frac{1}{2} \times (0,2,0,1,0,12)$ becomes

$$(0,1,0,\tfrac{1}{2},0,6)$$

$R_2^* = R_2 - 2R_1^*$ so $(3,2,0,0,1,18) - (0,2,0,1,0,12)$

yields $(3,0,0,-1,1,6)$

$R_3^* = R_3 + 5R_1^*$ so $(-3,-5,0,0,0,0) + (0,5,0,\frac{5}{2},0,30)$

yields $(-3,0,0,\frac{5}{2},0,30)$

now we can rewrite the tableau and check for optimality.

	X_1	X_2	X_3	X_4	X_5	RHS
R_0	1	0	1	0	0	4
R_1	0	1	0	1/2	0	6
R_2	$\boxed{3}$	0	0	-1	1	6
R_3	-3	0	0	$\frac{5}{2}$	0	30

Iteration 2

Since there is a negative number in bottom row, not optimal yet. Look for most negative value, then minimum ratio test to identify the pivot element. X_1 is entering basic variable (circle this column)

- $R_0 = 4/1 = 4$
- $R_1 = 6/0 = \infty$ (unbounded)
- $R_2 = 6/3 = 2$ ← minimum! hence X_5 is leaving basic variable (note how it has a single 1 and rest are 0s) X_2 can only be increased to 2!

On next page, write row operations for GJ elimination.

Let's start w/ the pivot element. ④

- $R_2^* = \frac{1}{3} R_2 \rightarrow \frac{1}{3}(3,0,0,-1,1,6) \rightarrow (1,0,0,-\frac{1}{3},\frac{1}{3},2)$
- $R_0^* = R_0 - R_2^* \rightarrow (1,0,1,0,0,4)-(1,0,0,-\frac{1}{3},\frac{1}{3},2) = (0,0,1,\frac{1}{3},\frac{-2}{3},2)$
- $R_3^* = R_3 + 3R_2^* \rightarrow (-3,0,0,\frac{5}{2},0,30)+(3,0,0,-1,1,6)$

yields $(0,0,0,\frac{3}{2},1,36)$

now we can write the revised tableau.

	X_1	X_2	X_3	X_4	X_5	RHS
R_0	0	0	1	1/3	-2/3	2
R_1	0	1	0	1/2	0	6
R_2	1	0	0	-1/3	1/3	2
R_3	0	0	0	3/2	1	36

no negative number in bottom row means optimal!

$Z = 36$ when $(X_1, X_2, X_3, X_4, X_5) = (2, 6, 2, 0, 0)$

remember how to read off the solution → if there is a column with one "1" and rest are "0", read off from RHS.
If column filled with junk (not one "1" and rest "0"), it's value is 0.